

Executive Summary

This assessment evaluates your capability to architect and deliver enterprise-grade AI solutions that integrates machine learning, **Generative AI (GenAI)**, and **Agentic AI** to solve complex business challenges

Objective

Design and implement a **comprehensive revenue optimization platform** that not only predicts no-shows but provides actionable intelligence across multiple business dimensions.

Note to Candidates Please be aware that this assessment is primarily evaluating your **capability to be creative and exploratory** while remaining **competent to build a simple prototype**. While the current scope involves a single dataset and specific tasks, you must pay close attention to **future scaling considerations** in your architectural and coding choices

Task 1: Advanced Business Intelligence & Exploratory Analytics

Candidates must perform an EDA using the noshow.db dataset within a Jupyter Notebook. The notebook should be structured to communicate findings to both technical and non-technical stakeholders.

Some suggestions [Candidates are open to explore as much as possible]:

- Multi-Dimensional Analysis such as cohort analysis across customer segments (first-time vs. returning, geography, platform)
- Execute customer lifetime value (CLV) estimation incorporating no-show probability
- Conduct A/B test simulation framework for policy interventions
- Perform survival analysis to model time-to-cancellation patterns
- Develop a risk scoring methodology that combines multiple factors
- Create an interactive executive dashboard with drill-down capabilities to automate insight generation with business context
- Prepare a strategic recommendation memo (max 2 pages) for C-suite consumption

Task 2: Enterprise-Grade ML Architecture & MLOps Pipeline

Build a production-ready, scalable ML platform with comprehensive MLOps integration.

Some suggestions [Candidates are open to explore as much as possible]:

- Implement and compare **at least 5 models**: Traditional ML, Deep Learning, Ensemble methods, custom loss functions, multi-task learning etc.
- Develop interaction features and polynomial transformations with feature selection
- Perform fairness audits across customer segments (demographic parity, equal opportunity)
- Conduct model interpretability analysis (SHAP, LIME, counterfactual explanations)
- Create data versioning and experiment tracking (MLflow, Weights & Biases, or DVC), model registry with version control and rollback capabilities
- Develop automated model retraining pipelines with drift detection
- Build CI/CD integration with automated testing (unit, integration, performance tests)
- Design for distributed processing (Dask, Ray, or PySpark), optimized for low-latency predictions (<100ms response time) and or create horizontal scaling architecture

Task 3: Advanced GenAI & Multi-Agent AI Orchestration

Demonstrate mastery of cutting-edge AI technologies to create autonomous, intelligent business systems.

Some suggestions [Candidates are open to explore as much as possible]:

- Implement **multi-modal LLM integration** (GPT-4, Claude, or open-source alternatives)
- Develop RAG (Retrieval-Augmented Generation) system for historical booking insights
- Create personalized intervention recommendations using LLMs
- Design and implement an **autonomous multi-agent framework** with specialized agents including inter-agent communication protocol and agent coordination logic with task prioritization
- Build agent memory systems for context retention
- Develop agent performance monitoring and evaluation
- Use LLMs to translate SHAP values into executive-friendly narratives
- Build conversational BI for interactive "What-If" scenario analysis tools

Task 4: Full-Stack Enterprise Application Development

Build a production-grade, user-centric application that demonstrates full-stack competency.

Some suggestions [Candidates are open to explore as much as possible]:

- Functional UI (React/TypeScript/MEAN) with interactive data visualization with drill-down capabilities
- Design RESTful API using FastAPI or Flask with health check and monitoring endpoints
- Implement authentication and authorization (JWT tokens)
- Create WebSocket connections for real-time updates with rate limiting and request throttling
- Implement comprehensive API documentation (OpenAPI/Swagger)
- Implement connection pooling and query optimization
- Create data access layer with ORM (SQLAlchemy)
- Build caching strategy (Redis) for frequent queries
- Containerize application using Docker/Docker Compose with multi-stage builds
- Implement environment-specific configurations

Submission Deliverables

A complete submission must include the following:

1. **src/ Folder:** Python modules for the ML, GenAI, and Agentic AI components.
2. **app.py:** The script for the front-end application.
3. **run.sh:** An executable bash script to run the modules (excluding dependency installation).
4. **requirements.txt:** A list of all necessary dependencies.
5. Multi-agent system architecture diagram showing data flow, model lifecycle, and deployment strategy
6. API documentation and Postman collection
7. **README.md:** A comprehensive document detailing the **pipeline design (with flowcharts)**, feature engineering processes, model evaluation metrics, and a **QA report**
8. Agent interaction protocols documentation
9. Evaluation report on agent performance and reliability

Submission Channel: Submit via GitHub Link

Evaluation Criteria

Candidates are assessed on their **clean coding practices**, their ability to build **scalable AI solutions**, and their skill in presenting **AI-driven business outcomes** to diverse audiences.

Data Fields Description List of Attributes Attribute	Description
booking_id	Unique customer booking ID
no_show	If the customer is a No-Show: 0 = Show, 1 = No-Show
branch	Hotel branch
booking_month	Month the booking was made by the customer
arrival_month	Month the customer plan to arrive at the hotel
arrival_day	Day date the customer plan to arrive at the hotel
checkout_month	Month the customer plan to checkout of the hotel
checkout_day	Day date the customer plan to checkout of the hotel
country	Nationality of the customer
first_time	If it is the first time customer staying in the hotel
room	Room type booked by the customer
price	Price of the room booked by the customer
platform	Platform used to book the room by the customer
num_adults	Number of adults staying
num_children	Number of children staying